INCH-POUND
MIL-C-5/1H
21 June 2011
SUPERSEDING
MIL-C-5/1G
15 December 2003

MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, MICA DIELECTRIC,
STYLE CM15

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-C-5.

INACTIVE FOR NEW DESIGN
after 31 March 1999

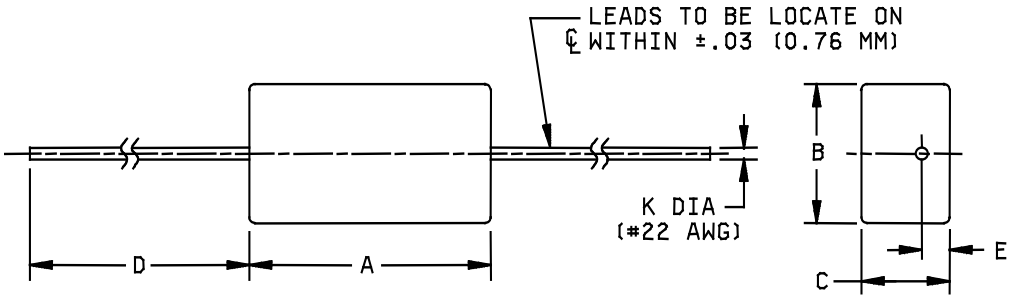


FIGURE 1

DIMENSIONS						
DIM	Inches			mm		
	MIN	NOM	MAX	MIN	NOM	MAX
A	.48	.52	.55	12.19	13.21	13.97
B	.27	.30	.31	6.86	7.62	7.87
C	.14	.19	.22	3.56	4.83	5.59
D	1.12	---	---	28.45	---	---
E	.06	---	---	1.52	---	---
K	.023	.025	.027	.58	.64	.69

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Approximate weight .05 ounce.
4. Unless otherwise specified, tolerance is ±.02.

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REQUIREMENTS:

Dimensions: See figure 1.

Capacitance value: See table I.

Capacitance tolerance: See table I.

Symbol	Capacitance tolerance
D	± 0.5 pF
F	± 1 percent
G	± 2 percent
J	± 5 percent

DC rated voltage: See table I.

Operating temperature range: See table I.

Symbol	Temperature range
N	-55°C to +85°C
O	-55°C to +125°C
P	-55°C to +150°C

Vibration grade: 3 (10 to 2,000 Hz).

Characteristic: See table I.

Symbol	Temperature coefficient P/M/°C	Capacitance drift
C	-200 to + 200	± (0.5 percent +0.1 PF)
E	-20 to + 100	± (0.1 percent +0.1 PF)
F	0 to + 70	± (0.05 percent +0.1 PF)

Barometric pressure (qualification only): As specified in MIL-C-5 with condition D (100,000 ft) and 150 percent of rated dc voltage.

Immersion: As specified in MIL-C-5 except, capacitance change shall not exceed ±2 percent of nominal value or 1 pf, whichever is greater.

Moisture resistance: As specified in MIL-C-5 except, capacitance change shall not exceed ±2 percent of nominal value or 1 pf, whichever is greater.

Qualification inspection: Qualification inspection is required. A manufacturer that is currently listed on the QPL for at least one other MIL-C-5 style may become qualified to the CM15 style by contacting the qualifying activity listed in the basic specification.

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TABLE I. Style CM15.

Type Designation 1/	Capacitance (pf)	DC working voltage (volts)	Operating temperature range	Characteristics available for capacitance tolerances			
				±.5 pf (D)	±1% (F)	±2% (G)	± 5% (J)
CM15CD010D-3	1	500	N,O,P	C			
CM15CD020D-3	2	500	N,O,P	C			
CM15CD030D-3	3	500	N,O,P	C			
CM15CD050D-3	5	500	N,O,P	C			
CM15CD100J-3	10	500	N,O,P				C
CM15CD110J-3	11	500	N,O,P				C
CM15CD120J-3	12	500	N,O,P				C
CM15CD130J-3	13	500	N,O,P				C
CM15CD150J-3	15	500	N,O,P				C
CM15CD160J-3	16	500	N,O,P				C
CM15CD180J-3	18	500	N,O,P				C
CM15CD200J-3	20	500	N,O,P				C
CM15CD220J-3	22	500	N,O,P				C
CM15CD240J-3	24	500	N,O,P				C
CM15-D270--3	27	500	N,O,P			C, E	C, E
CM15-D300--3	30	500	N,O,P			C, E	C, E
CM15-D330--3	33	500	N,O,P			C, E	C, E
CM15-D360--3	36	500	N,O,P			C, E	C, E
CM15-D390--3	39	500	N,O,P			C, E	C, E
CM15-D430--3	43	500	N,O,P			C, E	C, E
CM15-D470--3	47	500	N,O,P			C, E	C, E
CM15-D510--3	51	500	N,O,P		C, E, F	C, E, F	C, E, F
CM15-D560--3	56	500	N,O,P		C, E, F	C, E, F	C, E, F
CM15-D620--3	62	500	N,O,P		C, E, F	C, E, F	C, E, F
CM15-D680--3	68	500	N,O,P		C, E, F	C, E, F	C, E, F
CM15-D750--3	75	500	N,O,P		C, E, F	C, E, F	C, E, F
CM15-D820--3	82	500	N,O,P		C, E, F	C, E, F	C, E, F
CM15-D910--3	91	500	N,O,P		C, E, F	C, E, F	C, E, F
CM15-D101--3	100	500	N,O,P		C, E, F	C, E, F	C, E, F
CM15-D111--3	110	500	N,O,P		C, E, F	C, E, F	C, E, F
CM15-D121--3	120	500	N,O,P		C, E, F	C, E, F	C, E, F
CM15-D131--3	130	500	N,O,P		C, E, F	C, E, F	C, E, F
CM15-D151--3	150	500	N,O,P		C, E, F	C, E, F	C, E, F
CM15-D161--3	160	500	N,O,P		C, E, F	C, E, F	C, E, F
CM15-D181--3	180	500	N,O,P		C, E, F	C, E, F	C, E, F
CM15-D201--3	200	500	N,O,P		C, E, F	C, E, F	C, E, F
CM15-D221--3	220	500	N,O,P		C, E, F	C, E, F	C, E, F
CM15-D241--3	240	500	N,O,P		C, E, F	C, E, F	C, E, F
CM15-D271--3	270	500	N,O,P		C, E, F	C, E, F	C, E, F
CM15-D301--3	300	500	N,O,P		C, E, F	C, E, F	C, E, F
CM15-D331--3	330	500	N,O,P		C, E, F	C, E, F	C, E, F
CM15-D361--3	360	500	N,O,P		C, E, F	C, E, F	C, E, F
CM15-D391--3	390	500	N,O,P		C, E, F	C, E, F	C, E, F
CM15-D431--3	430	500	N,O,P		C, E, F	C, E, F	C, E, F
CM15-D471--3	470	500	N,O,P		C, E, F	C, E, F	C, E, F
CM15-D511--3	510	500	N,O,P		C, E, F	C, E, F	C, E, F

1/ Complete type designation, where applicable, will include additional symbols to indicate characteristic (C, E, or F), capacitance tolerance (D, F, G, or J), and temperature range (N, O, or P).

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Supersession data: Capacitors as described herein with capacitance values ranging from 51 through 510 pf, may be considered as suitable replacements for comparable items formerly covered by MIL-C-5/3C. Recommended replacements are as follows:

MIL-C-5/3C items	Recommended replacement items
CM25E510GN3 thru CM25E511GN3	CM15ED510GN3 thru CM15ED511GN3
CM25E510GO3 thru CM25E511GO3	CM15ED510GO3 thru CM15ED511GO3
CM25E510GP3 thru CM25E511GP3	CM15ED510GP3 thru CM15ED511GP3
CM25E510JN3 thru CM25E511JN3	CM15ED510JN3 thru CM15ED511JN3
CM25E510JO3 thru CM25E511JO3	CM15ED510JO3 thru CM15ED511JO3
CM25E510JP3 thru CM25E511JP3	CM15ED510JP3 thru CM15ED511JP3
CM25F910GN3 thru CM25F511GN3	CM15FD910GN3 thru CM15FD511GN3
CM25F910GO3 thru CM25F511GO3	CM15FD910GO3 thru CM15FD511GO3
CM25F910GP3 thru CM25F511GP3	CM15FD910GP3 thru CM15FD511GP3
CM25F910JN3 thru CM25F511JN3	CM15FD910JN3 thru CM15FD511JN3
CM25F910JO3 thru CM25F511JO3	CM15FD910JO3 thru CM15FD511JO3
CM25F910JP3 thru CM25F511JP3	CM15FD910JP3 thru CM15FD511JP3

CHANGES FROM PREVIOUS ISSUE. The margins of this specification are marked with asterisks to indicate where changes from previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5910-2011-027)

Review activities:
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Air Force - 99

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